

Question	Answer	Mark	Mark scheme	Additional guidance
19	$\frac{5}{3}$	P1	for process to derive an equation in x, eg $\frac{x}{4x-1} = \frac{6x+5}{12x+31}$	
		P1	for complete process to remove fractions, eg $x(12x+31) = (6x+5)(4x-1)$	Must be correct use of brackets
		P1	for process to reduce to a quadratic equation, eg $12x^2 - 17x - 5 = 0$	Award for correct LHS only.
		P1	for process to solve the quadratic equation by factorisation or use of quadratic formula, eg $(4x+1)(3x-5) = 0$	Award for correct LHS only. Accept substitution into the formula; $\frac{- -17 \pm \sqrt{(-17)^2 - 4 \times 12 \times -5}}{2 \times 12}$
		A1	for $\frac{5}{3}$ oe	Accept answers in the range 1.66 to 1.67 as equivalent